Which route would you choose?

You need to move 15 pallets of fragile electronics from Fremont to Austin. Your budget is $4,000.

OPTION A / LOWER COST

$2,100

Fremont > Phoenix > Austin

One transfer

OPTION B / EARLIER ARRIVAL

$3,400

Fremont > Austin, direct

Arrives 8 hours earlier

Before choosing, what else would you need to know?

Illustrative prices and services from the fixed synthetic scenario.

Meet RouteMind AI

A conversational assistant for comparing suitable ground-freight routes.

DESCRIBE

Tell us the shipment

Locations, quantity, budget, deadline and cargo-handling needs.

COMPARE

Understand the options

Review costs, arrival times, transfers and supporting information.

EXPLORE

Ask what changes

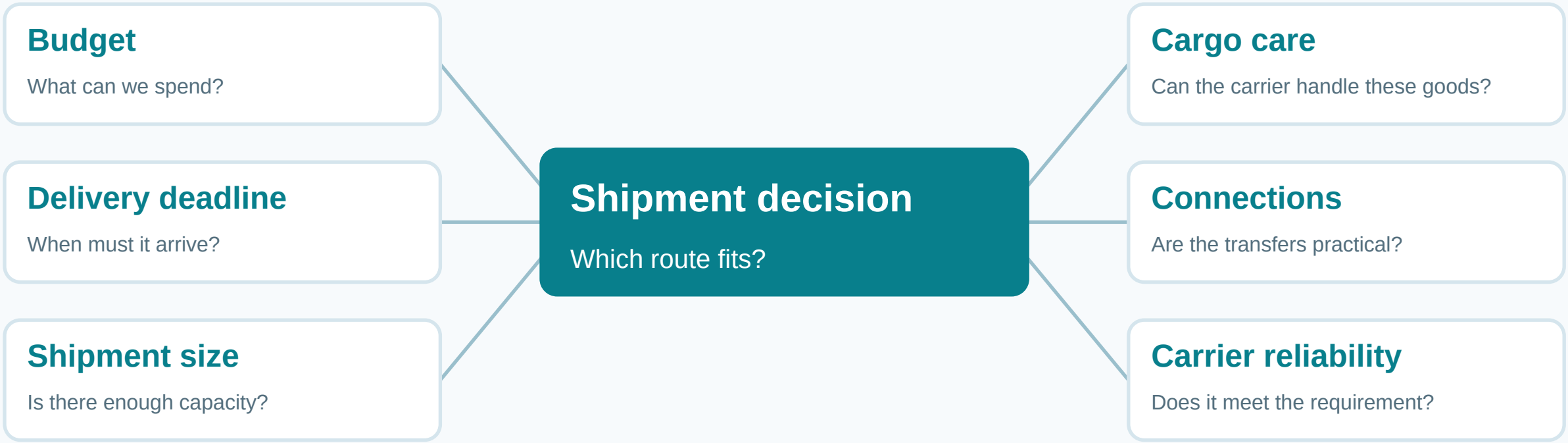
“Make it faster, but keep the budget.” Then review the trade-off.

Decision support: the user reviews the options and makes the final choice.

Project team: Chetana Patel | Dilara Siraeva | Minh N Phan

What makes a route suitable?

A route must satisfy the shipment’s requirements—not just offer a low price.



A change to one requirement can change the best route.

From a request to a decision you can explain

The challenge is to connect the user’s intent, operational constraints and supporting information.

How can we turn a shipment request into feasible route options, explain the trade-offs, and show the evidence behind the recommendation?

UNDERSTAND

Clarify the need

Ask about missing or ambiguous requirements before planning.

VERIFY

Check what is feasible

Respect budget, timing, capacity, connections and handling requirements.

EXPLAIN

Make evidence visible

Show the trade-offs and say when information is missing or conflicting.

Two kinds of information. Two different jobs.

SQL checks operational facts. Retrieval-augmented generation (RAG) connects answers to source passages.

STRUCTURED FACTS / SQL + OPTIMIZER

What services actually fit?

Prices, schedules, available pallet space and explicit handling flags.

Query the facts, then calculate and verify feasible routes.

WRITTEN INFORMATION / RAG

What do the documents say?

Carrier descriptions and handling guidance, expressed in varied language.

Find relevant passages and show their sources.

RAG supports document understanding.

It does not replace database checks or determine route eligibility.